BUSINESS SCHOOL

SKEMA

Corporate Finance
PGE M1 Fall 2026

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Session 5: Multiples

Housekeeping

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Mergers & Acquisitions

Mergers & Acquisitions + Add to myFT

Starbucks has explored takeover of Chipotle in restaurant megadeal

Purchase of $41bn market-value burrito chain would rank as largest restaurant acquisition of all time



Starbucks' chief executive Brian Niccol joined Starbucks from Chipotle in 2024 © Brendan McDermid/Reuters

What price should SBUX pay?

Three Approaches to Valuation



Intrinsic valuation. The value of an asset is a function of its fundamentals – cash flows, growth and risk.

- ▷ Discounted Cash Flow–WACC
- ▷ Adjusted Present Value (Not in CFM)

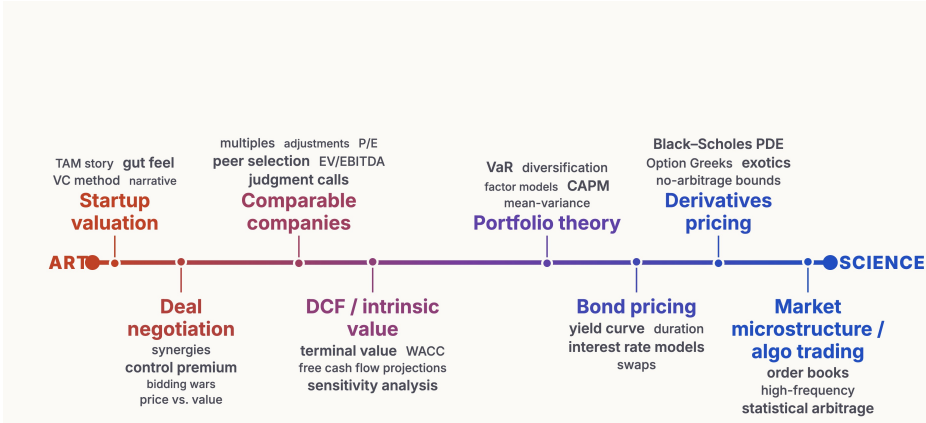
Relative valuation. The value of an asset is estimated based upon what investors are paying for similar assets.

- ▷ Multiples (today)

Contingent claim valuation. When the cash flows on an asset are contingent on an external event, the value can be estimated using option pricing models.

- ▷ Option pricing (Not in CFM)

Finance: Art vs Science



Who performs valuation – and why

Transactions	Advisory	Compliance
Investment banks	Management consultants	Auditors & accountants
Capital markets teams	Equity analysts	Courts & regulators
Private equity & VC	Corporate finance	



Most equity valuations on Wall Street are relative valuations

- Almost 85% of equity research reports are based on relative valuation
- More than 50% of all acquisition valuations are based on multiples

There are more DCF valuations in consulting and corporate finance, but they often incorporate some relative valuations.

- The terminal value in DCF valuations is sometimes estimated with a multiple

Transactions



- **Investment banks (M&A advisory)** – the target's bank builds a valuation (comps + DCF + precedent transactions) to argue the board should reject or demand more; the acquirer's bank builds one to justify the offer price and the premium over the undisturbed share price; a **fairness opinion** is the bank formally certifying to the board that the agreed price is fair from a financial point of view, often required before the board can approve the deal
- **Capital markets teams (ECM/DCM)** – in an IPO, equity capital markets bankers and the lead underwriters build a valuation range (comps-based, since there's no trading history yet) to set the offer price band shown on the roadshow; mispricing shows up immediately as a large first-day pop (money left on the table) or a broken syndicate (deal priced too high, stock falls below issue price)
- **Private equity and venture capital** – at entry, the deal team values the target to decide the maximum price to bid in an auction, usually via a leveraged buyout (LBO) model that works backward from a target IRR; at exit (sale or IPO), the same comps/DCF toolkit is used to size the realized return reported to the fund's limited partners

Advisory



- **Management consultants** – typically engaged for a specific strategic question: should the client enter a new geography, launch a new product line, or divest an underperforming division; the output is usually a multi-scenario DCF (base/upside/downside) comparing the net present value created under each path, feeding into a board-level recommendation rather than a transaction price
- **Equity research analysts (sell-side)** – cover a set of public stocks, publish a 12-month price target built from a blend of DCF and comps, and attach a buy/sell/hold rating; the price target is what moves when the analyst updates the model after each quarterly earnings release, and is read by portfolio managers deciding whether to buy or sell the stock
- **Corporate finance** / **FP&A teams (in-house)** – own the company's own long-range financial plan and apply it to capital budgeting (comparing a project's NPV against the firm's hurdle rate, usually close to WACC), to screening acquisition targets before bringing in a bank, and to deciding whether a business unit is worth more kept or sold

Compliance

- **Auditors and accountants** – under IFRS/US GAAP, goodwill and other assets must be tested at least annually for **impairment**: if the asset's fair value (estimated via DCF or comps) falls below its carrying value on the balance sheet, the company must write it down and take a charge to earnings; after an acquisition closes, auditors also perform **purchase price allocation** (PPA), valuing each acquired asset and liability individually to allocate the purchase price across the target's balance sheet
- **Courts and regulators** – in a **minority squeeze-out** (a controlling shareholder forcing out remaining minority holders), courts determine "fair value" per share, often triggering competing DCF valuations from each side's expert witness; tax authorities value private company shares (e.g., for estate tax or transfer pricing) where no market price exists; and antitrust regulators assess whether a merger's combined market power would harm competition, which can involve the same comps/DCF tools applied to post-merger market share and pricing power

Relative Valuation Recipe

The value of an asset is compared to the values assessed by the market for comparable assets

To do relative valuation, we need to:

i. Identify comparable assets and obtain market/book values for them.

ii. Standardize these numbers because the absolute values cannot be directly compared (firms can have many differences). Take averages/median of these standardized values. This is the multiple.

iii. Compare the multiple for the asset that is analyzed to the multiple for comparable assets, to assess whether the asset is under- or over-valued.

Price-to-earnings (P/E)



Definition: Share price ÷ EPS, equivalently Market Cap ÷ Net Income. Values the equity claim directly.

Calculation steps (CMG, FY2025):

i Market Cap = Price × Shares outstanding = \$37 × 1,304.4M = \$48,261M

ii P/E = Market Cap ÷ Net Income = \$48,261M ÷ \$1,535.8M = 31.4x

Pros

- Simple and intuitive – the most widely quoted multiple
- Captures the full income statement (interest, taxes, non-operating items)

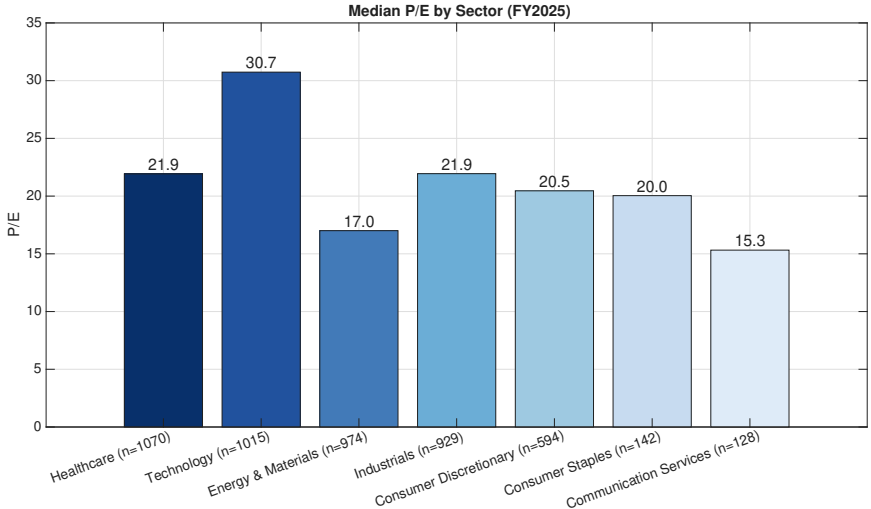
Cons

- Distorted by capital structure – leverage mechanically inflates EPS
- Meaningless or undefined when earnings are negative
- Sensitive to accounting choices and one-off items

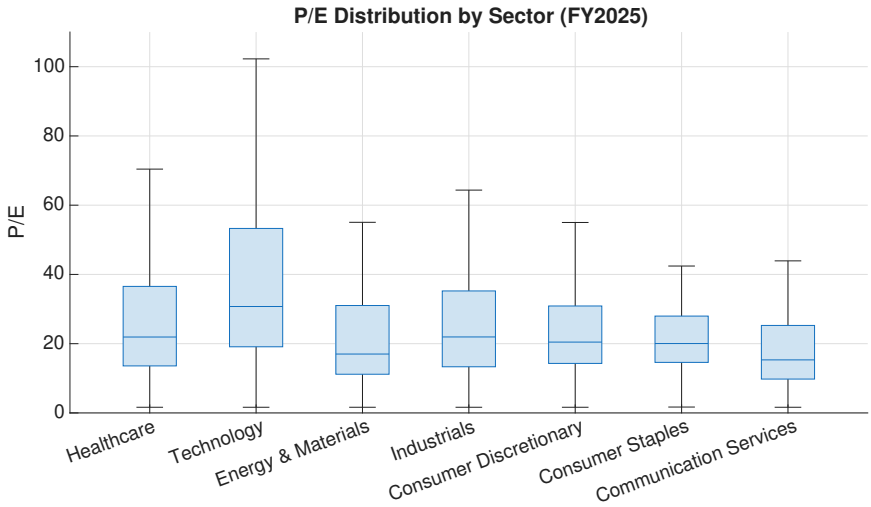
Putting Names to Faces

Sector	Top players
Technology	SAP, Nvidia, Apple, Microsoft, IBM
Healthcare	Novo Nordisk, Roche, AstraZeneca, Novartis, Sanofi
Energy & Materials	Shell, TotalEnergies, L'Oréal, Unilever, L'Air Liquide
Industrials	Air France-KLM, Caterpillar, Boeing, Accenture
Consumer Discretionary	LVMH, Toyota, Amazon, Tesla, McDonald's
Consumer Staples	Nestlé, Danone, Heineken, Philip Morris, Coca-Cola
Communication Services	Deutsche Telekom, Vodafone, Orange, Netflix, Disney

Median P/E ratio by Major Sector



P/E Ratio Distribution by Major Sector



Price-to-book (P/B)

Definition: Market Cap ÷ Book value of equity. Compares what the market pays to the accounting net worth on the balance sheet.

Calculation steps (CMG, FY2025):

i. Market Cap = \$37 × 1,304.4M = \$48,261M

ii. Book equity (from balance sheet) = \$2,831M

iii. P/B = \$48,261M ÷ \$2,831M = 17.0x

Pros

- Useful cross-check when earnings are volatile, negative, or non-meaningful
- Grounded in a tangible balance-sheet "floor," useful for asset-heavy or financial firms

Cons

- Can turn negative purely from debt-funded buybacks, unrelated to distress
- Least informative for intangible-heavy firms (brand value, internally developed R&D, skilled employees)

Textbook interpretation of P/B

P/B > 1. The market values the company's equity above its accounting net worth. The classic reading is that the market expects the firm to generate returns on equity above its cost of equity — i.e., it's creating value, and investors are willing to pay a premium over the liquidation/replacement-cost floor for that expected future profitability.

$$\text{ROE} > k$$

P/B < 1. The market thinks the company is destroying value, and investors would not pay full book value for a dollar of that equity.

$$\text{ROE} < k$$

where

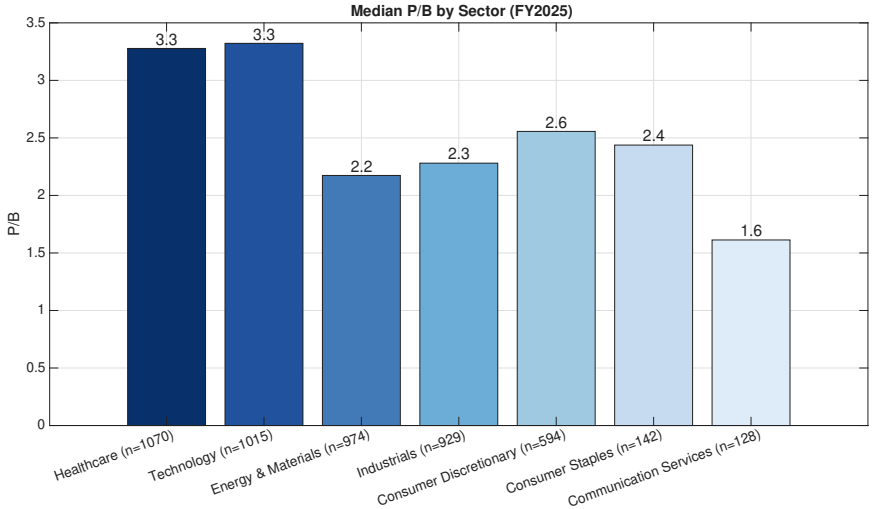
$$\text{ROE}_{t+1} = \frac{\text{Net Income}_{t+1}}{\text{Book Equity}_t}$$

(how much profit did a dollar of equity capital generate)

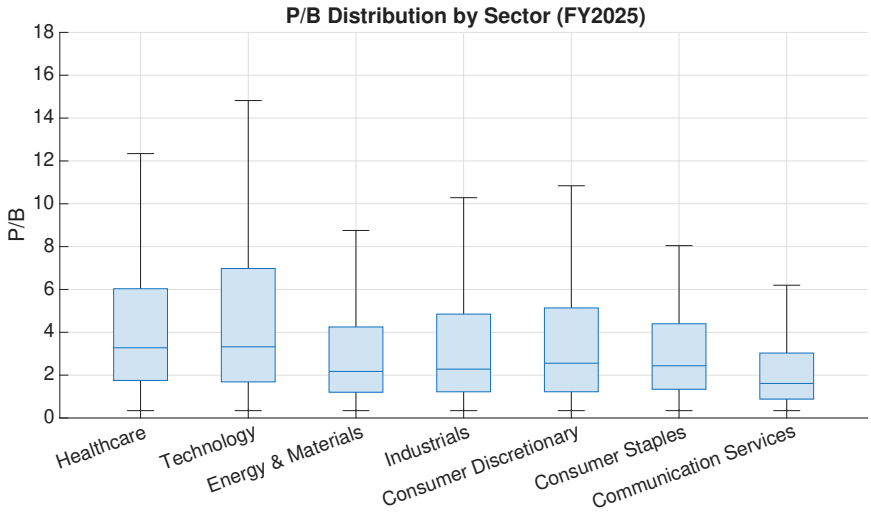
Derivation



Median P/B Ratio by Major Sector



P/B Ratio Distribution by Major Sector



EV/EBITDA

Definition: Enterprise Value ÷ EBITDA. Values the entire firm (debt + equity claims together), independent of capital structure.

Calculation steps (CMG, FY2025):

i. Market Cap = \$37 × 1,304.4M = \$48,261M

ii. EBITDA = EBIT + D&A = \$1,994M + \$361M = \$2,355M

iii. EV = Market Cap + Debt − Cash = \$48,261M + \$5,076M − \$1,049M = \$52,288M

iv. EV/EBITDA = \$52,288M ÷ \$2,355M = 22.2x

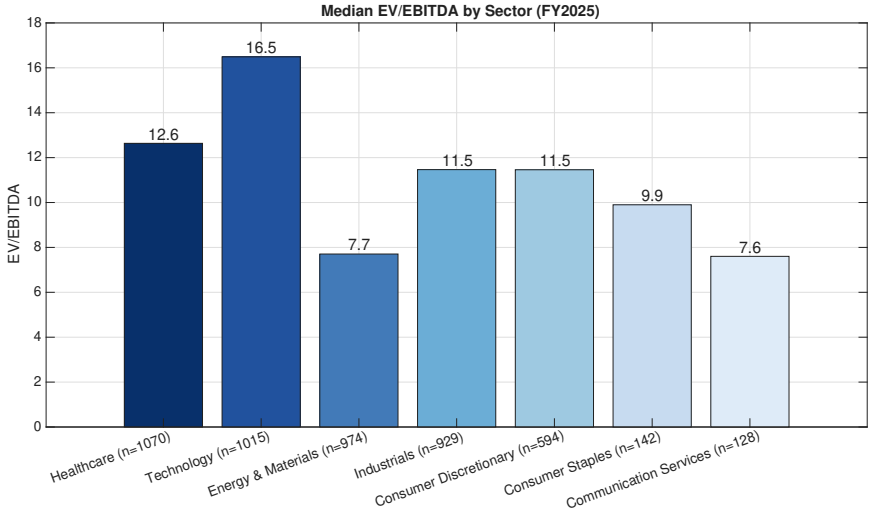
Pros

- Capital-structure neutral – comparable across firms with different leverage
- EBITDA is often used as a rough proxy for operating cash flow, stripping out non-cash D&A and financing/tax effects

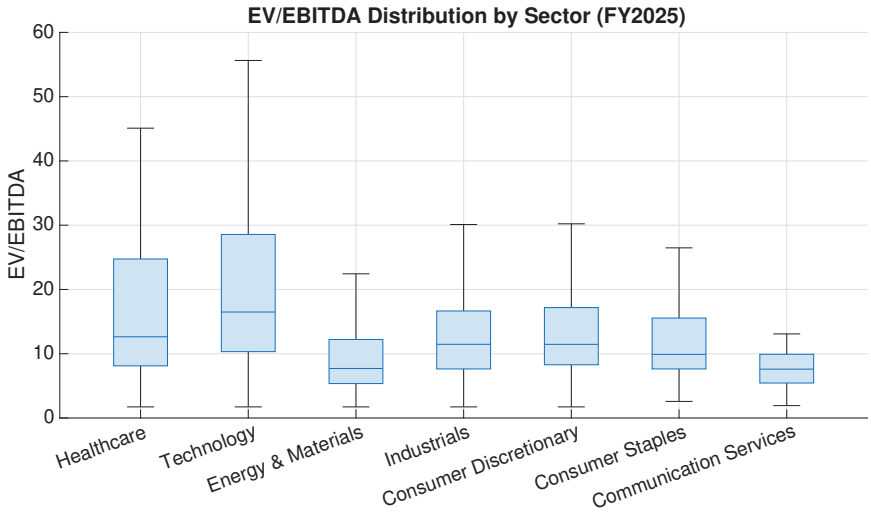
Cons

- ”EBITDA as a proxy for FCF” is only a rough approximation – ignores capex and working-capital needs
- Can overstate value for capital-intensive businesses still investing heavily in growth

Median EV/EBITDA by Major Sector



EV/EBITDA Distribution by Major Sector



EV/EBIT

Definition: Enterprise Value ÷ EBIT. Also capital-structure neutral, but after depreciation and amortization.

Calculation steps (CMG, FY2025):

i. EV = $52,288M (as above)

ii. EBIT (operating income) = $1,994M

iii. EV/EBIT = $52,288M ÷ $1,994M = 26.2x

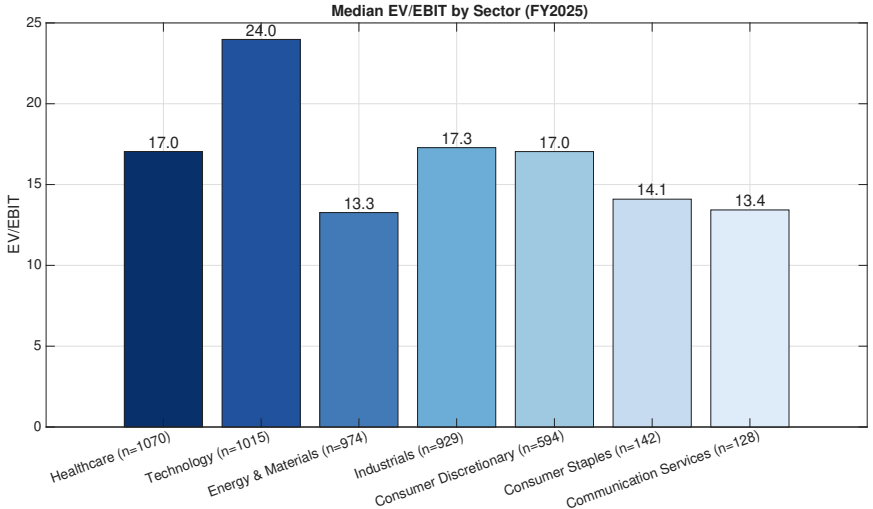
Pros

- Accounts for depreciation – better reflects the ongoing cost of maintaining the asset base than EBITDA
- Still capital-structure neutral, useful for cross-company comparison

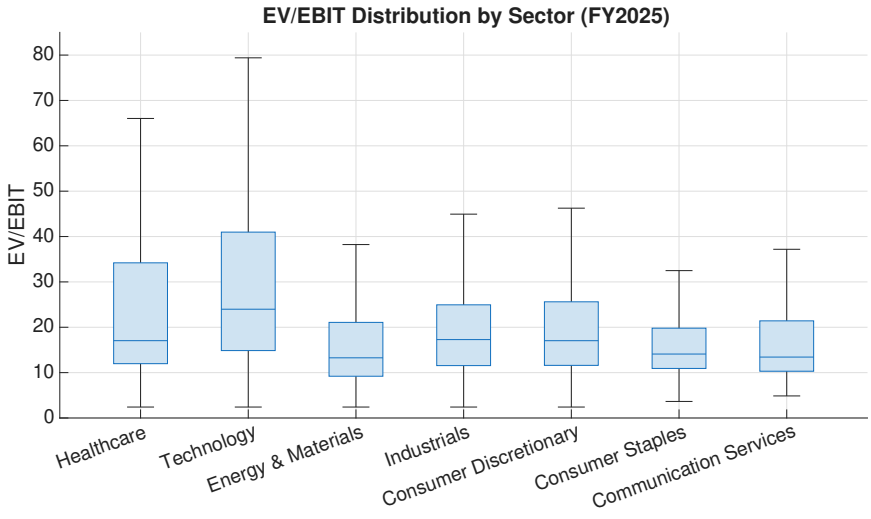
Cons

- Depreciation methods and useful-life assumptions vary by firm and accounting regime
- Like EV/EBITDA, doesn't directly capture capex timing or working-capital needs

Median EV/EBIT by Major Sector



EV/EBIT Distribution by Major Sector



Applying the P/E multiple: implied price

Idea. take a comparable company's multiple (not CMG's own) and apply it to CMG's own fundamentals to see what CMG *would be worth* if the market priced it like its peers.

Step 1 – EPS (CMG, FY2025):

$$\text{EPS} = \frac{\text{Net income}}{\text{Shares outstanding}} = \frac{\$1{,}535.8\text{M}}{1{,}304.4\text{M}} = \$1.18$$

Step 2 – apply the comp (Incumbents tier) median P/E:

$$\text{Implied price} = \text{Peer median P/E} \times \text{EPS} = 25.4 \times \$1.18 = \$29.97$$

Step 3 – compare to actual market price:

$$\text{CMG actual price (FY2025)} = \$37.00$$

If implied price $>$ actual price: CMG looks cheap relative to its peer group on this multiple. If implied price $<$ actual price: CMG looks expensive – the market is already paying a premium versus peers.

Who are the competitors?



“The fast-casual, quick-service, and casual dining segments of the restaurant industry are highly competitive with respect to, among other things, taste, price, food quality and presentation, customer service, location, convenience, brand reputation, and cleanliness and ambience of each restaurant. Our competition includes a variety of restaurants in each of these segments, including locally-owned restaurants, as well as national and regional chains. Competition from food delivery services, which offer meals from a wide variety of restaurants, also has increased in recent years and is expected to continue to increase. Many of our competitors also offer dine-in, carry-out, online, catering, and delivery services. Among our main competitors are restaurant formats that claim to serve higher quality ingredients without artificial flavors, colors and preservatives, and that serve food quickly and at a reasonable price.”

Chipotle Annual 10-K filing, Item 1

Comps: Young, Fast-growth

Ticker	Mkt Cap ($M)	EV ($M)	P/E	P/B	EV/EBITDA	EV/EBIT
CAVA	6,815	6,889	106.9×	8.7×	53.1×	122.8×
WING	6,568	7,610	37.7×	n.m.	35.0×	39.6×
SHAK	3,267	3,809	71.5×	6.2×	22.0×	57.2×
SG	801	1,066	n.m.	2.2×	n.m.	n.m.
BROS	7,778	8,598	97.4×	11.4×	30.2×	50.7×
Average			78.4×	7.2×	35.1×	67.6×
Median			84.4×	7.5×	32.6×	54.0×
CMG	48,261	52,288	31.4×	17.0×	22.2×	26.2×

Comps: Established players / incumbents



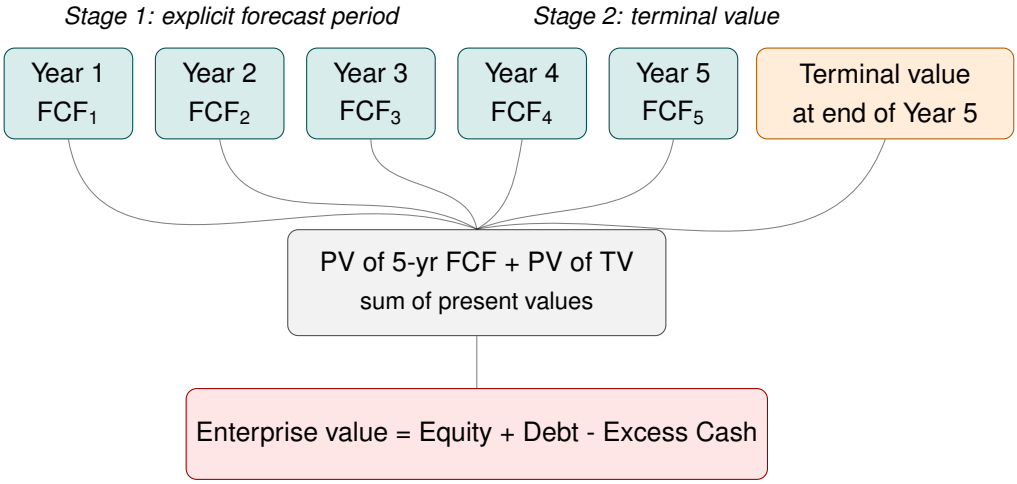
Ticker	Mkt Cap ($M)	EV ($M)	P/E	P/B	EV/EBITDA	EV/EBIT
MCD	217,181	271,220	25.4×	n.m.	19.4×	21.9×
QSR	23,630	38,146	30.5×	6.5×	13.7×	15.2×
YUM	41,905	54,193	26.9×	n.m.	18.4×	20.1×
DPZ	14,017	18,723	23.3×	n.m.	17.9×	19.6×
WEN	1,585	5,390	9.6×	13.5×	11.4×	16.8×
Average			23.1×	10.0×	16.2×	18.7×
Median			25.4×	10.0×	17.9×	19.6×
CMG	48,261	52,288	31.4×	17.0×	22.2×	26.2×

Comps: Broader sector (SIC 5812, all eating places)

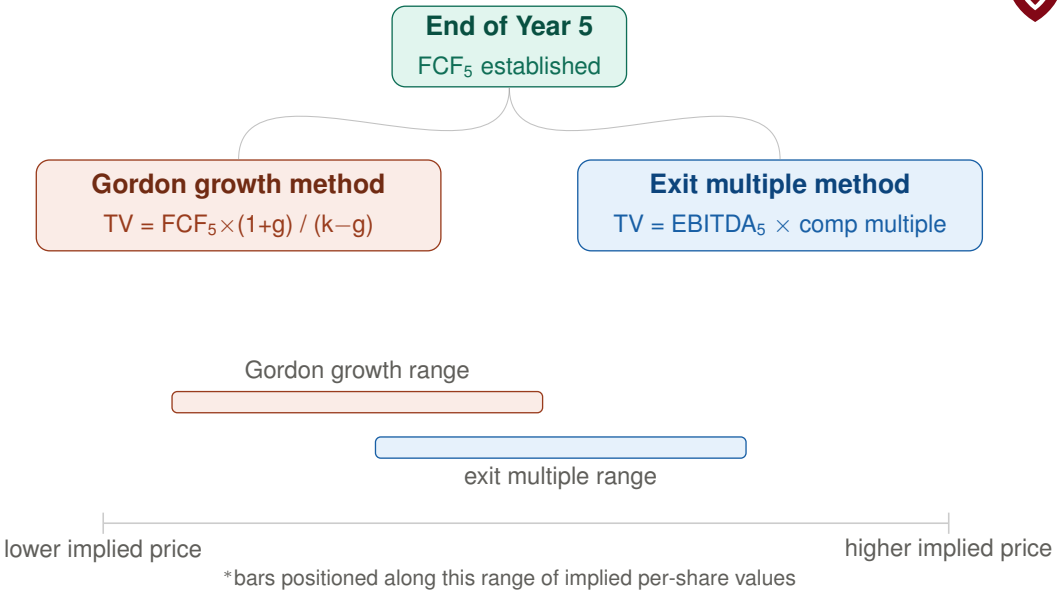
	P/E	P/B	EV/EBITDA	EV/EBIT
Average	32.3×	4.7×	22.6×	28.3×
Median	24.3×	2.6×	13.3×	19.9×
CMG	31.4×	17.0×	22.2×	26.2×

44 comps across the full SIC 5812 universe (excl. CMG) — unfiltered beyond removing negative/not-meaningful denominators.

Combined intrinsic valuation + multiples approach



*All cash flows discounted by the company's cost of capital


End of Year 5
FCF5 established
Gordon growth method
TV = FCF5×(1+g) / (k−g)
Exit multiple method
TV = EBITDA5 × comp multiple
Gordon growth range
exit multiple range
lower implied price
higher implied price
*bars positioned along this range of implied per-share values

Why cross-check terminal value?

Terminal value dominates total enterprise value

$\approx$ 81% of DCF-implied EV comes from the terminal value
(only $\approx$ 19% from the 5 explicit forecast years)

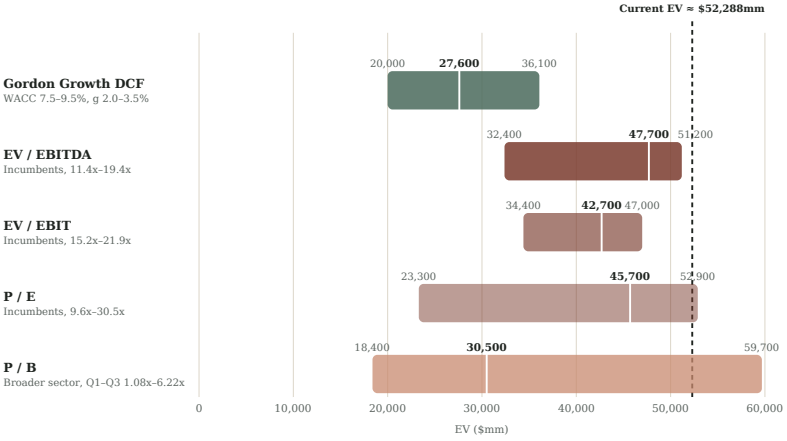
And TV is highly sensitive to two long-run assumptions (illustrative base case: WACC 7.9%, g 3.0%)

	+1pp	−1pp
WACC	−17% TV	+26% TV
Terminal growth g	+26% TV	−17% TV

DCF assumptions

- Revenue base (FY2025): $11,926M
- Revenue growth: 8.0% $\rightarrow$ 7.5% $\rightarrow$ 7.0% $\rightarrow$ 6.5% $\rightarrow$ 6.0% (Years 1–5)
  - Tapering toward mature same-store-sales plus unit growth
- EBITDA margin: 19.5% $\rightarrow$ 19.8% $\rightarrow$ 20.0% $\rightarrow$ 20.3% $\rightarrow$ 20.5%
  - Modest operating leverage as new units mature
- Cost of capital: WACC $\approx$ 7.9%
- Terminal value: cross-checked via Gordon growth ($g = 3.0\%$) and exit multiple (15x EV/EBITDA)

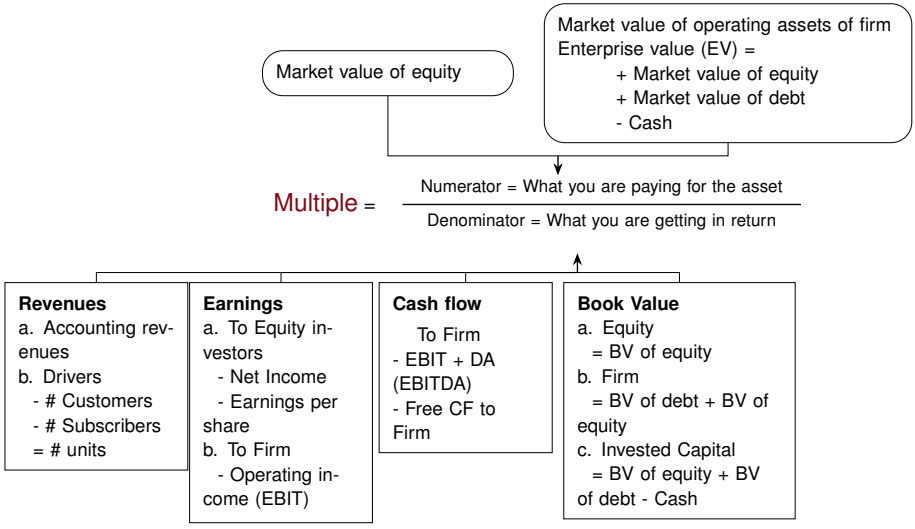
CMG Valuation Football Field



P/E basis: terminal value is computed on equity, then converted to EV using current net debt (~$4.0B) for comparability.

P/B basis: projected Year-5 book equity of $11,745mm (current book equity rolled forward, full earnings retention, no dividends/buybacks), multiplied by broader-sector P/B quartiles (n=35, positive book equity only), then converted to EV using current net debt.

Choosing a Multiple



Choosing a Multiple

Is the multiple consistently defined?

- Important: The asset value (numerator) and the standardizing flow variable (denominator) should refer to the same claimholders in the firm (e.g., equity/debt/all investors)
- Example: Are these good multiples?

$$\frac{EV}{EBITDA} \qquad \frac{Price}{EBITDA} \qquad \frac{Price}{Earning}$$

Is the multiple uniformly estimated?

- The variables used to define the multiple should be estimated uniformly across assets in the list of comparable firms
- Accounting rules to measure the numerator and denominator should be applied consistently across firms, assets, and countries (e.g., for earnings-based multiples)

Concerns

There are many valid multiples that can be used to value a company. Which multiple should we choose?

→ Criterion: *how is value measured and created?*

The answer is usually sector-specific.

- Retail: focus is on same store sales (turnover) and profit margins → revenue multiples
- Banks: cannot separate investment (loans) and financing decisions (deposits). Emphasis is on equity value, not enterprise value → price/book ratios dominate
- Technology: growth has been dominant theme → P/EG ratios were invented in this sector

Limitations

1. Comparability: even within industry, many differences between similar firms are not captured by the multiples and will result in wrong valuations
   - ▷ Sub-divisions of the company
   - ▷ Expected future growth rates
   - ▷ Profitability (margins)
   - ▷ Risk (cost of capital)

2. Multiples depend on capital structure
   - ▷ Use P/E multiple when buying stock (no debt), which entitles shareholder to cash flows
   - ▷ Use enterprise value multiples if compare firms with different leverage

3. Entire industry may be over- or undervalued: if everyone is overpaying, you will too
   - ▷ Internet/AI boom vs. fast-growing firms with initially negative P/E (Tesla)

Summary of Multiples



Strengths

- Quick and simple
- You are not alone (can help “ride bubbles” in short term)

Weaknesses

- Comparable firms? Accounting measures? Ignored factors?
- Backward-looking, while DCF and APV are forward-looking
- If the market is overpaying, you will too

Conclusion

- Don’t rely solely on multiples!
- The multiples approach complements intrinsic valuation
  - Sanity check
  - Terminal value

Who Pays Dividends?

As of FY 2025

Sector	*N*	% Pay Dividend	% Buyback	% Either
Consumer Discretionary	594	39.4	65.8	72.7
Consumer Staples	142	52.1	56.3	71.8
Communication Services	128	42.2	57.0	65.6
Industrials	929	39.4	60.5	67.5
Technology	1015	17.3	56.8	60.3
Energy & Materials	974	29.0	36.2	43.5
Healthcare	1070	6.7	33.8	35.0
Total (pooled)	**4,852**	**25.9**	**49.4**	**54.7**

Dividends vs Buybacks

The Gordon-growth model says

$$\text{price} = \frac{\text{Dividends}}{k - g}$$

but most firms do not pay dividends.

- A firm's dividend policy is "sticky". It is extremely costly to lower dividends afterwards; markets heavily punish the stock price and it usually means something is wrong with the company.
- In contrast, share buybacks are more flexible and are considered discretionary without commitment.